



NCCER Module 22102



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Heavy Equipment Safety





Section One

Work Zone Safety

Objective

1. Explain work zone safety requirements.
 - a. Describe signs, barriers, and other devices and methods used for work zone safety.
 - b. Describe traffic control safety methods.



Performance Tasks

Under the supervision of an instructor, trainees should be able to do the following:

1. Place barricades and temporary traffic control devices for a highway construction zone.
2. Demonstrate how to use flags or paddles to control traffic.



1.0.0 Work Zone Safety

Accidents can be avoided by properly using signs, barricades, and other devices. Construction work zones are often in public areas.



1.1.0 Signs, Barriers, and Other Safety Devices

Unauthorized people and vehicles must be kept away from the work area. Operators should always be aware of state and local regulations.



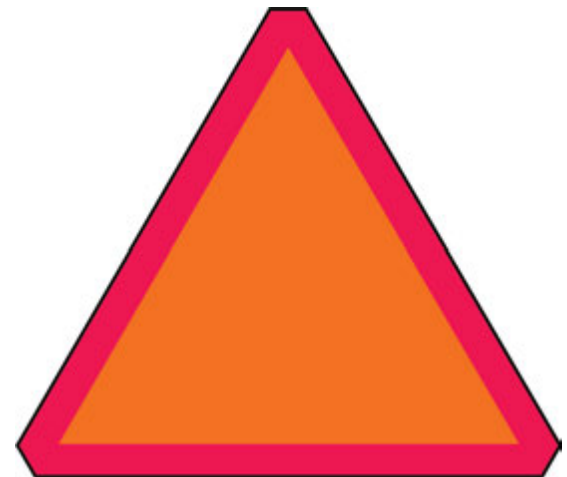
1.1.1 Signs (1 of 3)

- Danger signs are used to inform workers of immediate hazards.
- Caution signs are used to inform workers of potential hazards or unsafe conditions.



1.1.1 Signs (2 of 3)

- **Temporary traffic control (TTC)** signs redirect traffic and provide information to drivers and pedestrians.
- A slow-moving-vehicle emblem is used on vehicles that move at speeds of 25 miles per hour (mph) or less.



1.1.1 Signs (3 of 3)

Safety alert symbols: Danger, Warning, or Caution.

	⚠ DANGER Turning shaft will kill you or crush arm or leg. Stay away.		⚠ WARNING Explosion possible. Serious injury or equipment damage could occur. Follow directions carefully.		⚠ WARNING Looking into fiber-optic cable could result in permanent vision damage. Do not look into ends of fiber-optic or unidentified cable.	
	⚠ DANGER Electric shock. Contacting electric lines will cause death or serious injury. Know location of lines and stay away.			⚠ WARNING Fluid or air pressure could pierce skin and cause injury or death. Stay away.		⚠ WARNING Fire or explosion possible. Fumes could ignite and cause burns. No smoking, no flame, no spark.
	⚠ DANGER Deadly gases. Lack of oxygen or presence of gas will cause sickness or death. Provide ventilation.			⚠ WARNING Job-site hazards could cause death or serious injury. Use correct equipment and work methods. Use and maintain proper safety equipment.		⚠ WARNING Moving traffic - hazardous situation. Death or serious injury could result. Avoid moving vehicles, wear high visibility clothing, post appropriate warning signs.
	⚠ DANGER Moving parts. Being struck by wrench will kill or injure. Do not use drilling unit to turn or move drill string when wrench is used.			⚠ WARNING Crushing weight could cause death or serious injury. Use proper procedures and equipment or stay away.		⚠ WARNING Hot pressurized cooling system fluid could cause serious burns. Allow to cool before servicing.
				⚠ WARNING Moving parts could cut off hand or foot. Stay away.		⚠ WARNING Improper control function could cause death or serious injury. If control does not work as described in instructions, stop machine and have it serviced.

1.1.2 Safety Tags

Accident prevention tags are used as a temporary warning, like when a piece of equipment is broken.

START-UP INSTRUCTIONS

EQUIPMENT ID _____
LOCATION _____

Slip No.	Instructions

(A)

DO NOT OPERATE

Signed by _____
Date _____

(B)

OUT OF ORDER

Signed by _____
Date _____

(C)

AUTHORIZED PERSONNEL ONLY

OPERATION OF THIS EQUIPMENT IS RESTRICTED TO THE FOLLOWING PERSONS

NAME _____

Signed by _____
Date _____

(D)

DANGER

LIVE WIRE

SIGNED BY _____
DATE _____

(E)

DANGER

DO NOT REMOVE THIS TAG

REMARKS _____

SEE OTHER SIDE

(F)

1.1.3 Lockout Devices

- Lockout devices prevent someone from using broken equipment.
- They can have padlocks or electronic keypads.

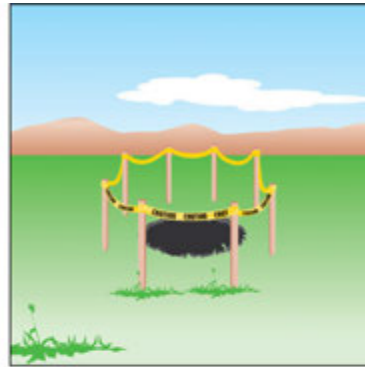


1.1.4 Barricades and Barriers

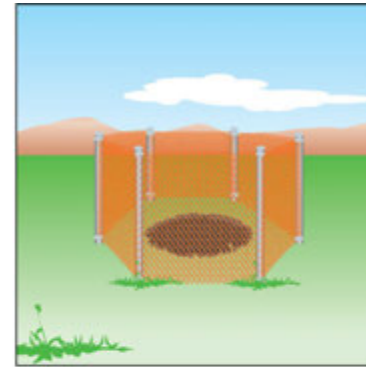
Each work zone may have different policies and procedures for how and when to use them.



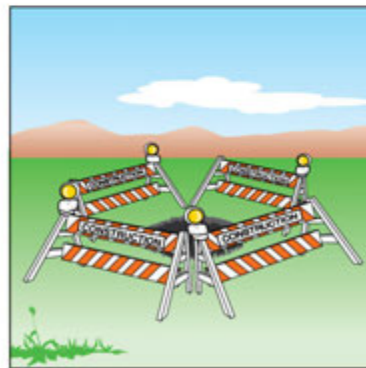
(A)



(B)



(C)



(D)



(E)

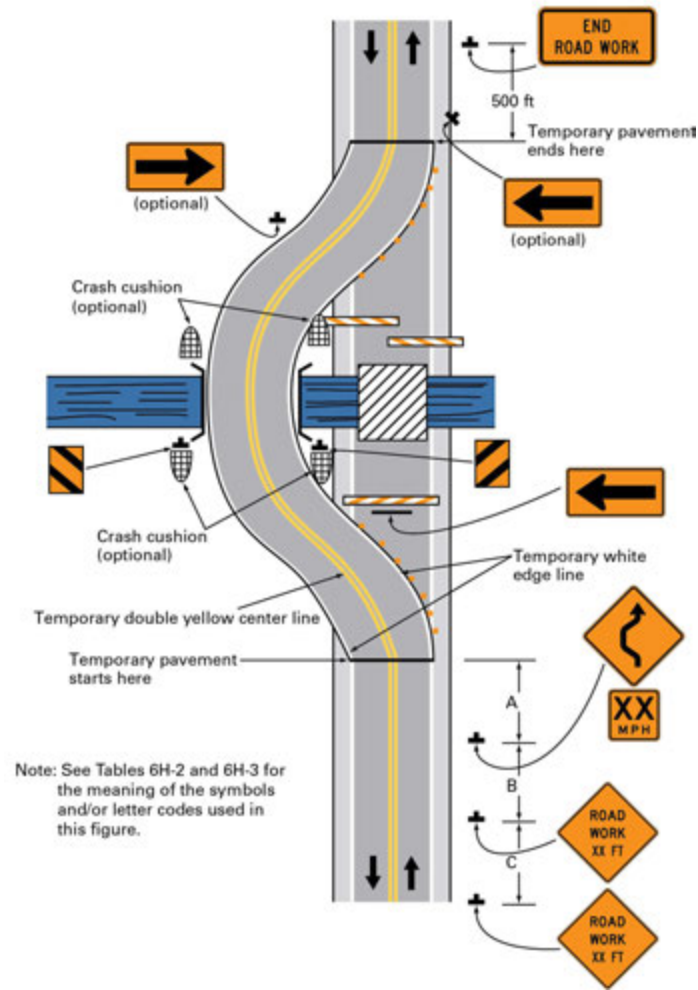
1.1.5 Audible and Hand Signals

- Audible signals such as alarms, bells, buzzers, whistles, and horns can be used to communicate hazards to workers.
- Hand signals are necessary when the surrounding noise levels are too loud to hear audible safety signals.
- Hand signals may differ across companies and from state to state. Always confirm which signals to use with your coworkers.



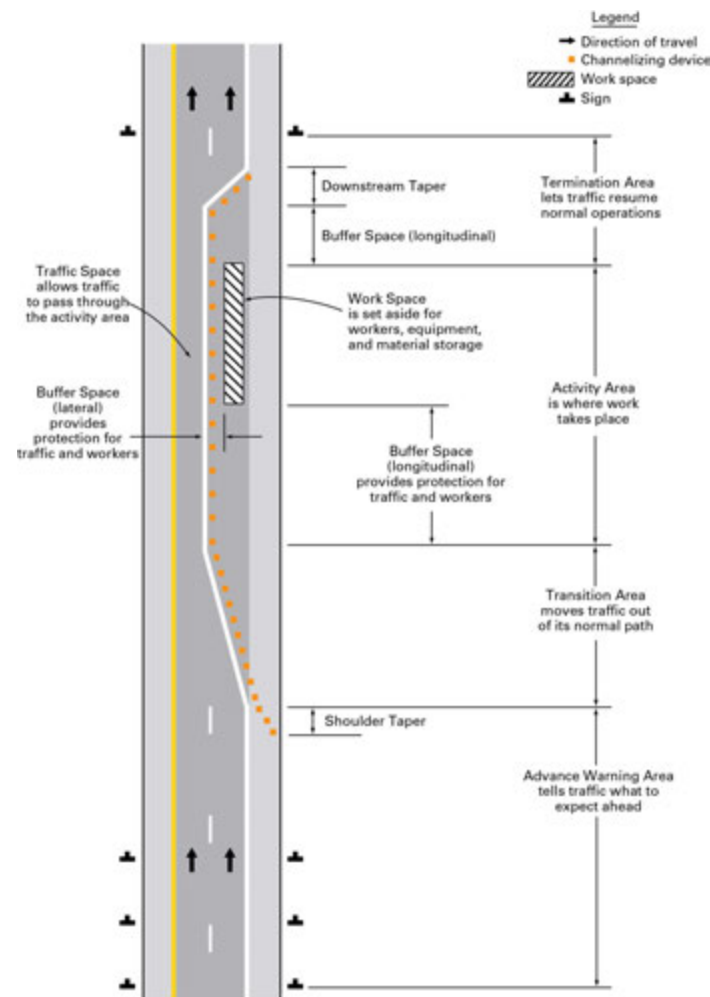
1.2.0 Traffic Control Safety

- The Manual on Uniform Traffic Control Devices (MUTCD) for Streets and Highways states the basic principles for traffic flow design.
- Specially trained workers create traffic flow plans.



1.2.1 Temporary Traffic Control Zones (1 of 4)

You will not be in charge of planning the TTC, but you do need to understand the layout and functions of the TTC to work safely.



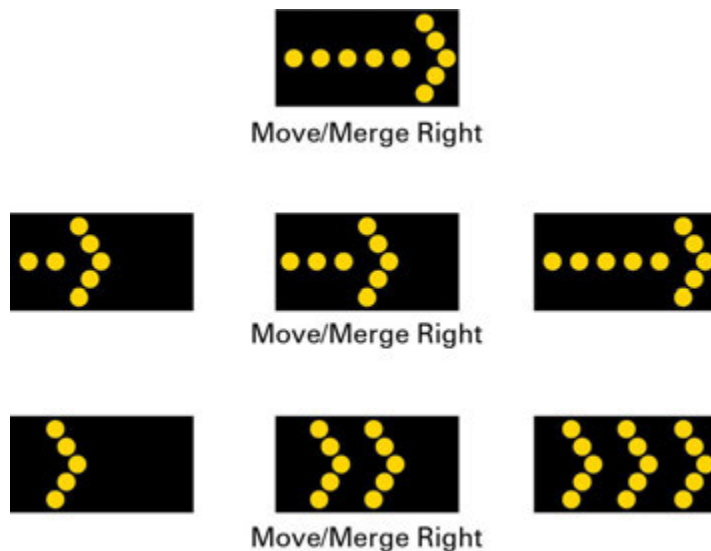
1.2.1 Temporary Traffic Control Zones (2 of 4)

The first part of the TTC zone is the advance warning area. Warning signs appear to let motorists know what to expect ahead.



1.2.1 Temporary Traffic Control Zones (3 of 4)

The transition area is used to move traffic into the desired path. This area has flashing, lighted signs or orange and white markers called **channeling devices**.



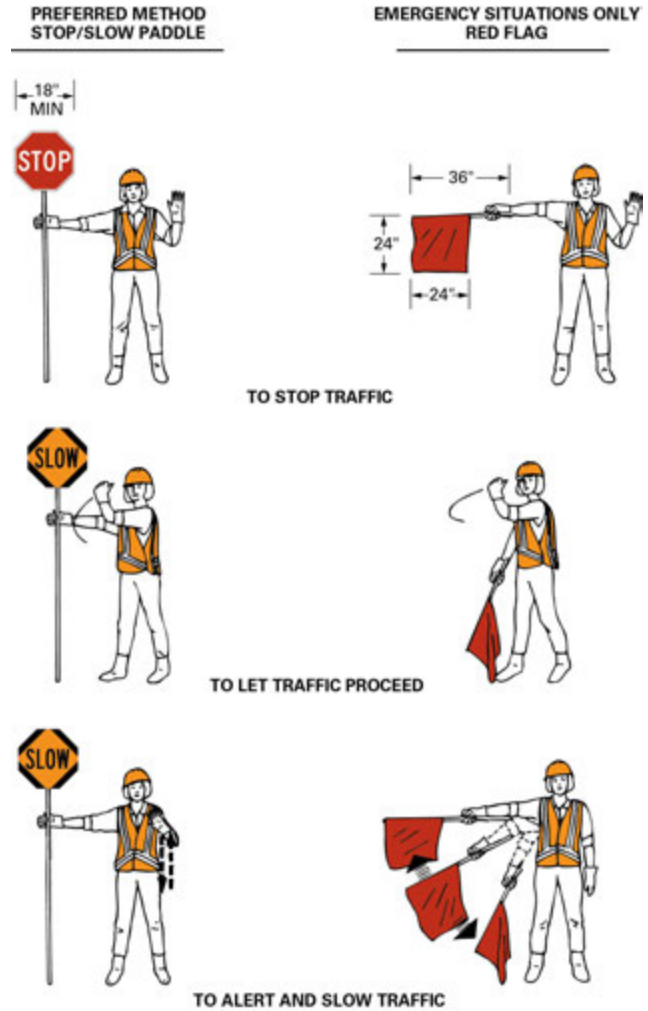
1.2.1 Temporary Traffic Control Zones (4 of 4)

- The longitudinal buffer zone separates the work area from traffic.
- Buffers must create a space in case a vehicle crosses through the barriers.
- There may be a worker assigned to monitor the buffer zones.



1.2.2 Flagger Control

- **Flaggers** must wear **retroreflective** material.
- A flagger must never step in front of a vehicle to stop it.
- Be aware that drivers may not understand signals.



1.2.3 Personal Safety

- Remain alert and stay aware of your surroundings at all times.
- Heavy equipment operators can stay alert by taking every opportunity to get out of the operator's seat.
- Workers on foot need to stay in view of anyone who is operating equipment.



Wrap Up – Trade Terms

Channeling device: Cones, barrels, or markers that are used to move traffic from its normal lane to a desired lane.

Flagger: A person who is specially trained to direct traffic through and around a work zone.

Retroreflective: A device or surface that reflects light with very little scattering.

Temporary traffic control (TTC): Device and plan used to safely divert traffic when the normal use of the road is disrupted due to constriction.



1.0.0 Section Review Question 1

1. The type of safety sign that is usually yellow with black letters is the _____.
 - a. warning sign
 - b. temporary traffic control sign
 - c. caution sign
 - d. safety alert



1.0.0 Section Review Question 1 Answer

1. The type of safety sign that is usually yellow with black letters is the _____.
 - a. warning sign
 - b. temporary traffic control sign
 - c. caution sign**
 - d. safety alert



1.0.0 Section Review Question 2

2. Information on how to design and use traffic control devices is provided in the _____.
- a. MUTCD
 - b. OSHA handbook
 - c. SDS
 - d. TTC handbook



1.0.0 Section Review Question 2 Answer

2. Information on how to design and use traffic control devices is provided in the _____.
- a. **MUTCD**
 - b. OSHA handbook
 - c. SDS
 - d. TTC handbook



Next...

Section 2.0.0
Hazard Communication



*Read Sections 2.0.0 through 2.3.0 and
complete the Section Review.*